



The Weather Analysis

Machine Learning for Weather Insights

Presented by

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Goal and Objectives

1

Goal

Analyze weather data using machine learning.

2

Objectives

Develop a pipeline for preprocessing weather datasets.

3

Objectives

Train models to forecast weather attributes.

4

Objectives

Reveal hidden patterns in weather features.

Introduction & Problem

Introduction

Weather prediction impacts agriculture and disaster management.

This project enhances weather pattern understanding.

Problem

Weather data contains complex interdependencies.

Predicting precipitation requires advanced data preprocessing.

2	Seep	153	166	327	1460	347	1195	1390	367	3,352	343
3	Soco	257	231	381	147%	440	2105	2190	388	3,15%	724
24	Soep	100	286	349	1490	347	3161	2197	257	3,573	560
34	Speo	390	231	590	1347	347	2672	2390	368	0,251	235
35	Speo	357	394	381	1455	347	3275	3590	335	0,50%	360
40	Spco	166	151	320	660	380	5575	3640	175	0,831	550
56	Spco	255	205	257	265	440	2275	2506	260	0,665	255

Dataset Description

Source

Kaggle, NOAA, or custom dataset.

Rows & Columns

95,936 rows and 17 columns.

Key Features

Temperature, Humidity, Wind Speed, Visibility, Pressure.

Data Preprocessing

1

Missing Values

Addressed missing values in categorical features.

2

Standardized

Standardized numerical features.

3

Encoded

Encoded categorical variables.

4

Extracted

Extracted temporal features.

DATA CLEANING



Model Selection & Training

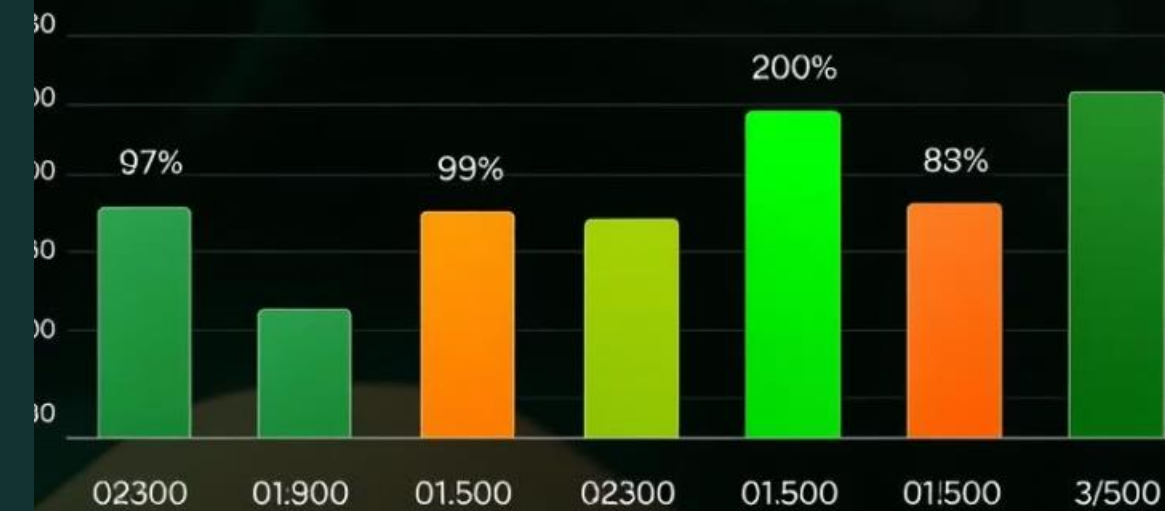
 Linear Regression

 Random Forest

 KMeans Clustering

Split data into training (80%) and testing (20%) subsets.

Machine Learning





Results & Achievements

0.99

Linear Regression

R² score

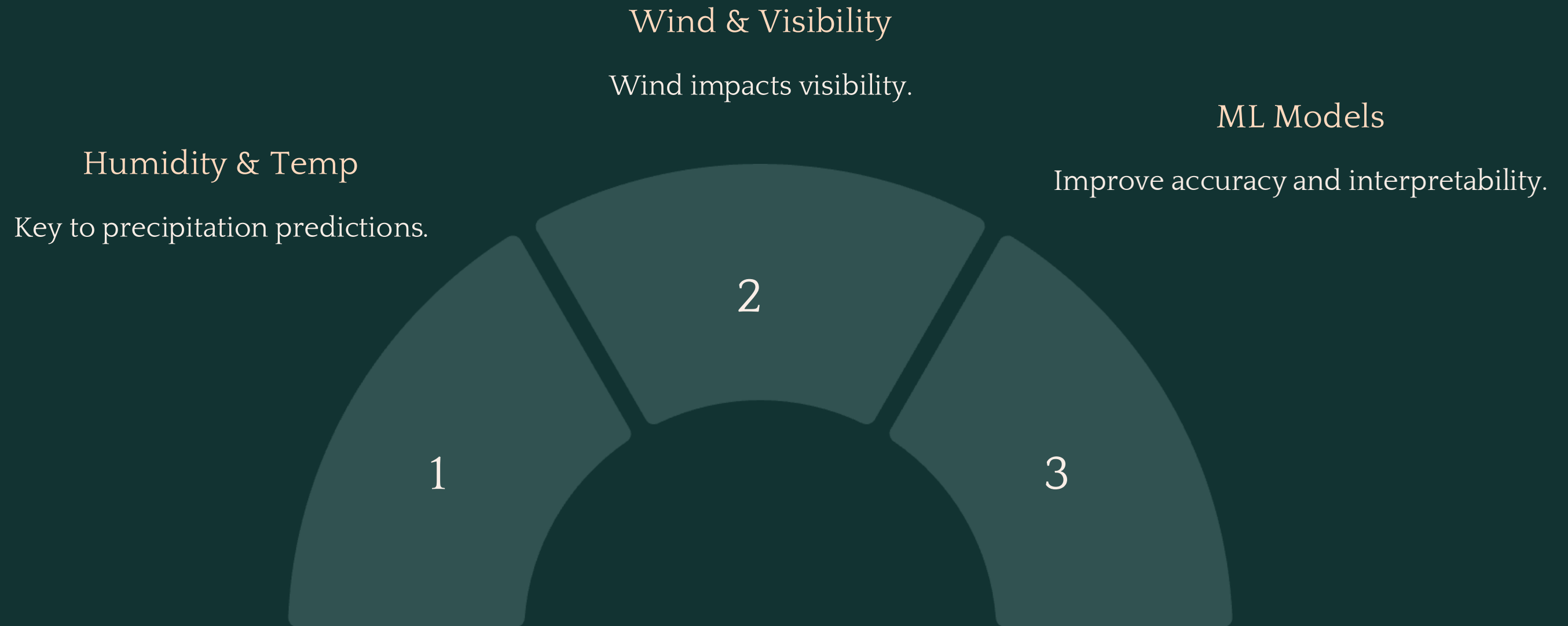
0.63

Random Forest

Accuracy

Improved feature representations through clustering.

Insights & Conclusion





Improvements & Future

Larger Datasets

Collect broader seasonal data.

Optimize Features

Reduce dimensionality.

Deep Learning

Explore LSTMs for real-time prediction.

github

SOCIAL CODING



GITHUB: -[Weather Analysis Github Link](#)